

User Interfaces

Inclusive Design

Inclusive Design aims to include as wide a range of users as possible while keeping design complexity low. It often involves added technology such as audio, hyperlinks, and alternative payment methods. However, achieving true inclusivity requires careful planning. Adapting existing designs after the fact can result in patchy, expensive designs. Focusing solely on HCI interaction paradigms for assistive techniques may limit the scope to interaction only, and following guidelines without context can be too theoretical.

Steps for Inclusive Design

- Adaptation must be “**designed into**” the system rather than implemented afterwards.
- Requires understanding the **global execution context** of a task.
- Breaks away from the traditional perspective of “typical” users interacting.
- Should be **non-platform specific**.
- Broaden the range of graphical toolkits (Stephanidis, 2000).

Suggested Guidelines

- **User characteristics:** Consider the level of expertise the target audience has in the subject area.
 - **Domain requirements:** What is the purpose of the application? What tasks are to be performed?
 - **Technology requirements:** Browsing and assistive technology.
 - **Pre-existing alternatives:** What other ways already exist to provide access (multimedia clips, telephone information lines)?
 - **Quality of alternatives:** What other ways could be created to provide alternative routes to the same goals? What are the accessibility barriers? (Sloan et al., 2004)
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Interaction Styles

Interaction styles are different approaches to how a user communicates with a system. The main types are:

- **Command Line** – e.g., DOS, network commands
- **Menus** – structured choices

- **Natural Language** – interpreting human language with different nuances
- **Question/Answer style** – query language to construct a query
- **Form Fill-in** – data capturing in labeled fields
- **WIMP** – Windows, Icons, Menus, and Pointing devices

Menus

Menus can be **single** (binary, multiple item, multiple selection, pull-down and pop-up, scrolling and 2-dimensional, alpha sliders, embedded links) or **multiple** (linear sequence, tree, acyclic, cyclic).

Menu Structures

- **Linear sequences and multiple menus** – A series of menus presented one after another.
- **Tree-structured menus** – Hierarchical navigation where users drill down from a root.
- **Acyclic** – Menus that allow multiple paths but no cycles.
- **Cyclic** – Menus that can loop back to previous options.

Menu Layout Guidelines

- **Titles:** For single menus, use a descriptive title. For linear sequences, titles should accurately represent stages. For tree-structured menus, the root must be clearly stated (e.g., MAIN). Use the same words in high-level menu items as in titles for next lower-level menus. Be consistent with placement of titles.
 - **Phrasing of menu items:** Use familiar and consistent terminology. Ensure items are distinct from one another. Use consistent and concise phrasing. Bring the keyword to the left.
 - **Consistency guidelines:**
 - Titles: centred or left justified
 - Item placement: left, may use blank lines or numbers between items
 - Instructions: identical, same position
 - Error messages: consistent position, terminology, and syntax
 - Status reports: consistent position and structure
 - **Progression:** Successive menus can be cascades, walking, or transparent menus. Walking menus can present a motor challenge. Transparent menus should not be higher than 50% opacity.
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Interaction Devices

- **Keyboards** – QWERTY layout, audio feedback
- **Pointing Devices:**
 - **Direct:** light pen, touch screen
 - **Indirect:** mouse, trackball, joystick, touchpad
- **Touch Screen:** Issues include smudges, hand obscuring view, and accuracy concerns.
- **Speech Recognition and Generation** – discrete-word and continuous-speech recognition.
- **Gaming Devices** – specialized controllers.

Speech Interaction

Discrete-word recognition involves individual words spoken by a specific person. It is used when the speaker's hands are busy, mobility is required, eyes are occupied, or in harsh/cramped conditions. **Background noise** influences accuracy.

Continuous-speech recognition allows dictating letters, but is limited.

Speech store and forward – messages stored and forwarded (e.g., weather, airlines, voice mail, audio tours).

Speech generation – used when the message is simple, short, will not be referred to later, deals with events in time, requires immediate response, or visual display is unavailable.

Interface Feedback

Error Messages

Why do users make errors? Lack of knowledge, incorrect understanding, inadvertent slips. Errors lead to feelings of inadequacy, confusion, and anxiety. An imperious tone condemning the user increases anxiety, causing further errors. Too generic messages do not provide enough information to correct the error. Obscure messages (e.g., FAC RJCT 0004444) offer no assistance.

Guidelines for Error Messages

- Be **specific** about what caused the error (e.g., "Type first letter: Send, Read, or Drop").
- Use **constructive guidance** and a positive tone.

- Use **user-centered phrasing**: instead of “ENTER DATA,” use “READY TO ACCEPT DATA.”
- **Appropriate physical format**: Reserve capital letters for serious warnings; avoid error codes; place messages appropriately.
- Develop effective error messages by **involving users**.
- Provide **multiple levels** of messages.
- Use **consistent grammatical form, terminology**, and abbreviations.
- Use **consistent visual format and placement**.

Colour in Feedback

- Place colour coding under user control.
- Consider the needs of **colour-deficient users**.
- Limit the number of colours to **no more than 4**.
- Use colour as a coding technique (e.g., all addresses one colour, labels another, data-entry fields another).
- Ensure colour coding supports the task (e.g., active window different colour, change colour when status changes).
- Consider **common expectations** about colour codes (e.g., warning messages in red – but note cultural differences).

Non-Anthropomorphic Design

- Avoid presenting computers as people.
- Choose appropriate humans for introductions and guidelines (culturally and gender sensitive).
- Use caution with computer-generated human faces or cartoon characters.
- Design **comprehensible, predictable, and controllable** interfaces.
- Provide user-centered overviews for orientation and closure.
- Do not use “I” when the computer responds to human actions; use “you” or state facts.

Data Display

- Keep **complexity low**.
- Present data in **directly usable form**.
- **Order lists** logically.

- **Label each page** in multi-page displays.
 - Maintain a **consistent format**.
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Response Times

Response Time is the number of seconds from when the user initiates an activity until the computer begins to present results. **User Think Time** is the number of seconds during which users think before entering the next action. Users plan while reading results, typing, and while the computer generates results or retrieves data. Precise measurement of User Think Time is difficult.

Factors Influencing Optimum Speed

- **Experience:** Novices may need slower response times; experienced users require rapid action.
- **Penalty for errors:** If low penalty, users prefer to work faster.
- **Task:** Typing, frequent tasks, data capturing, problem-solving tasks require different response times.
- **Expectations and attitude:** Past experiences and varied response times for the same task promote agitation.
- **Delay tolerance:** Varies by personality, age, cost, mood, culture, pressure, etc.
- **Task complexity:** More complex tasks allow longer response times.

Variability

- Users can adapt to variable situations; some take advantage of it.
- Modest variations ($\pm 50\%$) are tolerable with little effect on performance.
- Extreme changes cause frustration, though users still complete tasks.

Response Time Guidelines

- Users prefer **shorter** response times.
- Longer response times (>15 seconds) are **disruptive**.
- Users change usage profile with response time.
- Shorter response time leads to **shorter think time**.
- Faster pace may increase productivity but may increase error rates.
- Error recovery ease and time influence optimal response time.
- Response time should be **appropriate for the task**:

- Typing, cursor motion, mouse selection: **50–150 milliseconds**
 - Simple frequent tasks: **1 second**
 - Common tasks: **2–4 seconds**
 - Complex tasks: **8–12 seconds**
 - Users should be **advised of long delays**.
 - Modest variability is acceptable; unexpected delays may be disruptive.
 - Empirical tests help set suitable response times.
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Interface Types and Modalities

Multimedia

Multimedia combines different media in the same interface: text, graphics, video, sound, animation. Benefits include overcoming weaknesses of individual media, improving user experience (engagement and pleasure), and improving learning and retention. We remember:

- Only **10%** of what we see
- **20%** of what we hear
- **80%** of what we see, hear, and do

Disadvantages: Difficult to design integration; can be distractive (e.g., users focus on video and miss text).

Sound in Interfaces

Why Use Sound vs. Visual

- People cannot see information outside their view (eye focuses on a small area) but can hear all around them.
- We turn to the source of sound.
- User can choose not to look but it is harder to avoid hearing.
- **Reaction to sound is faster** than visual (auditory ~150ms, visual ~200ms, pain ~700ms).
- **Design implication:** Use sound to grab user attention.

Main problem with sound: It is temporal (lasts only a short time).

Speech Sound

Involves spoken words. Used for:

- Input (e.g., dialing a contact by voice)
- Output (e.g., error messages, feedback)
- Speech recognition in word processors, page scanners, web readers, home entertainment

Disadvantages:

- **Slow** because serial – must be listened to in full
- Many words needed to convey a simple message
- Can irritate people nearby
- Meaning depends on context (prior knowledge, accuracy of pronunciation)
- **Ambiguity** – system mishearing or misunderstanding (e.g., “add,” “hard,” “heard,” “herd”)

Non-Speech Sound

Not based on spoken language (whistling, humming). Used as input (e.g., whistling to take a picture) or output (e.g., notifications, ringtones).

Benefit: Fast – does not express meaning, therefore shorter.

Disadvantage: Meaningless – user must learn what sound stands for.

Direct Manipulation

Basic Principles

1. **Continuous representations** of objects and actions using meaningful visual metaphors.
2. **Physical actions** (e.g., turn buttons to increase volume).
3. **Rapid, incremental, reversible actions** – effects visible immediately; easy to recover.

Example: Drag and drop a folder – visual representation, feedback, user can undo.

Attributes of Direct Manipulation Systems

- Novices can learn basic functionality quickly.
- Experts can work quickly.
- Easy to remember operational concepts.
- Error messages are rarely needed.
- Users experience less anxiety because interface is comprehensible and actions can be reversed.

- Users gain a sense of confidence and mastery.
- **Multi-touch** allows new actions (e.g., two-finger zoom in/out) that are intuitive.

Multimodal User Interfaces

Modality is the sense by which a message is sent or received. A message is conveyed by a **medium** and received by a **modality**. Humans naturally combine two or more modalities (e.g., sight and hearing). The most widely combined modalities in interfaces are **vision and sound**.

Benefits

- Different ways of interaction
- Flexibility and efficiency
- Reduced errors and easy recovery
- Accommodates a wide range of users, contexts, applications
- Adapts seamlessly to changing environmental conditions
- Avoids overuse of one specific modality

Challenges

- Relies on recognizing user behaviour (handwriting, speech, gestures, eye movement)
- Inputs are uncertain, recognition is probabilistic, meaning is ambiguous

Why Combine Visuals and Sound?

- **Orientation:** Sight and hearing are interdependent; we turn to look at sources of sound.
- **Reduce load on visual system:** Most interfaces are visual-intensive; use hearing to reduce visual load.
- **Reduce information on screen:** Present information in sound to free screen space.
- **Reduce demand on visual attention:** Use sound to reduce visual attention (e.g., audio cues in driving).
- **Use sound to grab attention** (users cannot ignore sound).

Gestural Interfaces

Gesture is a motion of the body that contains information (e.g., waving goodbye). Gestural interfaces are designed to respond to natural gestures. A gesture consists of two parts: the gesture itself and **gesture recognition**.

Important Issues

1. Understanding the role of gesture in human communication.
2. Detecting gestures using tracking technologies.

Applications

- Home: gesturing with smartphone/tablet
- Entertainment: gaming
- Work and learning: collaborative tabletop, wall interfaces, surface interfaces (curved, fabric)

Research Directions

- Full body interaction
- Convergence with 3D, voice recognition, virtual reality

Benefits

- Based on natural human behaviour, therefore more intuitive
- More flexible
- Less visible hardware
- Improved user experience

Challenges

- High volumes of data input
- Reliance on visual and physical
- Gesture ambiguity and fatigue

Shareable User Interfaces

Designed for **more than one person** to use simultaneously. Includes interactive tabletops, walls, etc. Advantages: support flexible group working (joint browsing, sharing, manipulation). Examples: tabletop shared screen, ImmerseBoard for two collocated users.

Tangible User Interfaces

Allows a user to interact with a **virtual object** by manipulating a **physical object**. The physical object (real object) is mapped onto the digital object; changes to the physical object change the digital object.

Underlying MCRpd Model

- **Control**

- **Model**
- **View** composed of Rep-p (Physical Representation) and Rep-d (Digital Representation)

Integrates direct manipulation, 3D interfaces, seamless interaction, multimodal, gestural, augmented reality.

Four Basic Types (Garber, 2012)

1. **Interactive surface:** tangible objects placed and manipulated on flat interfaces; system interprets position, proximity.
2. **Tabletop interfaces:** combine multi-touch and tangible interaction.
3. **Constructive-assembly:** user attaches modular elements; system interprets space, order.
4. **Token and constraints:** uses tokens and structures that limit how users place and move objects.

Benefits

- Leverages natural ability to manipulate physical objects.
- Users are both physically and cognitively involved.
- Multiuser support improves social interaction.
- Does not demand as much visual attention as GUI.

Example

A tangible interface for molecular biology: user manipulates a physical model; camera captures data; software displays digital equivalent.

Evaluation Methods

Introduction to Evaluation

Evaluation tests usability and functionality of a system. It occurs in the laboratory, field, or in collaboration with users. It should be considered at all stages in the design life cycle.

Goals of Evaluation

- Assess extent of system functionality
- Assess effect of interface on user
- Identify specific problems

Selecting an Evaluation Method: Factors to Consider

1. **What to evaluate:** Whole UI, part (interaction, aesthetics), task (functionality), human performance (user behaviour).
2. **Why (purpose):** Determine usability for user groups, identify problems, compare design choices, determine impact on users.
3. **When:** During or after design, early or late.
4. **Where:** Natural context (field) or unnatural context (laboratory).
5. **Who:** Experts (domain, evaluation, UI design) or users.
6. **How:** Based on the above, choose appropriate method(s); sometimes methods are combined.

Types of Evaluation

1. **Formative vs. Summative:**
 - **Formative:** Early stages; refines ideas (helps “form” the product).
 - **Summative:** Later stages; decision-making (e.g., select optimal design).
2. **Qualitative vs. Quantitative:**
 - **Qualitative:** Detailed, non-numeric information.
 - **Quantitative:** Numeric data (e.g., measures of user behaviour); allows comparison.
3. **Analytical vs. Empirical:**
 - **Analytical:** Based on reasoning and analysis.
 - **Empirical:** Based on observations and measurements; involves real users.
4. **Expert vs. Participant-based:**
 - **Expert review (Inspection):** Only experts.
 - **Participant-based:** Involves users.
5. **Field vs. Experiment:**
 - **Experiment:** Manipulation of variables.
 - **Non-experiment:** No manipulation (survey, interview, observation).
6. **Formal models:** Formulas to calculate measures.
7. **Automated software evaluation.**

Heuristic Evaluation

Proposed by Nielsen and Molich. Also called **discount evaluation** – quick and cheap. May be used at any stage (paper prototypes to finished UI). Involves **3–5 experts**.

Procedure

1. Experts examine UI using design guidelines (e.g., Nielsen's heuristics) to identify violations.
2. Each identifies problems independently and makes notes.
3. Problems are aggregated into a final list.
4. Each expert rates each problem by **severity** (scale 0–4):
 - **0**: Not a problem
 - **1**: Cosmetic problem
 - **2**: Minor problem
 - **3**: Major problem
 - **4**: Catastrophic problem
5. Average score of all experts represents problem severity.

Severity determined by: **frequency of occurrence, impact on user, persistence**.

Advantages

- Cost-effective: identifies most problems quickly and at low cost.
- Few practical and ethical issues.
- Requires external expert (you don't evaluate your own design).
- Experts knowledgeable of domain and users.

Common Problems

- **Expert bias**
- Severe problems may be missed
- **False positives** – many trivial problems identified
- Lacks theoretical assumptions

Cognitive Walkthrough

Proposed by Polson et al. Focuses on **learnability** – how well the design supports the user in learning the task. Experts play the role of first-time users or novices, assuming their thoughts and actions.

Input

- User profile (characteristics and capabilities)
- Scenario, description of interface (e.g., screens and correct sequence based on HTA)

Procedure

Experts walk through the interface asking four questions (based on Norman's stages of action model); any negative response signifies a problem:

1. **Will users try to achieve the correct outcome?** (Understand the goal)
2. **Will users notice the correct action to achieve the goal?** (Visibility of correct buttons, etc.)
3. **Will users recognize this action as the correct one?** (Match between goal and labels/buttons – clear and distinct)
4. **Will users understand the feedback?** (Interpret progress toward outcome)

Advantages

- Structured
- Based on theory (theory of exploratory learning)
- Cost-effective

Problems

- May not find all problems (use with other methods)
- May find problems that do not exist or that users overcome by commonsense

Experiments

Controlled experiments provide evidence from data to support claims (hypotheses). They are used to evaluate UI design issues. Two main types:

- **Absolute:** Evaluate a specific UI on its own.
- **Comparative:** Compare two or more designs.

Typically conducted in a controlled environment (laboratory). **Basic assumption:** cause and effect.

Procedure

1. Identify problem or purpose.
2. Formulate hypothesis (e.g., "Colour coding improves accuracy of selection").
3. Design and conduct experiment to collect data.
4. Analyze data for evidence to **accept or reject** hypothesis.

Key Concepts

- **Independent Variable (IV):** Controlled or manipulated variable.
- **Dependent Variable (DV):** Variable that changes due to IV.
- Select sample of users and divide into groups:
 - **Experiment group:** Experiences the IV.
 - **Control group:** Does not experience the IV.
- Groups can be **independent** (different users) or **within-subject** (same group does both conditions).
- Compare results **between groups** or **within groups**.

Example

- Hypothesis: "Colour coding improves accurate selection."
- IV: Colour; DV: accurate selection.
- Experiment group: colour coding provided; control group: no colour coding.
- Collect data on errors; compare; accept or reject hypothesis.

Advantages

- Collects quantitative, objective data.
- Provides evidence to support claims.
- Allows comparison of alternate designs.

Common Problems

- Evidence may not be conclusive due to **confounding variables** (distraction, motivation, etc.).
- Variations in human abilities; sample may not represent entire user population.
- Ethical issues.

Field Methods

Involve data gathering from observations and measurements. Each method has strengths and weaknesses; use **triangulation**. Broadly divided into **experiment** and **non-experiment**.

Methods Covered (Brief)

- **Observation** (including Thinkaloud – self-study)
- **Interview**

- **Survey/Questionnaire**
- **Ethnography**
- **Contextual Enquiry**
- **Experiment** (covered above)

Analytics (Software Tools)

Software tools may be used for evaluation, locally or remotely. They log user activities to understand how the UI is used, user behaviour, and preferences.

Example: Tools like TechSmith's **Morae** can measure mouse clicks per task over time, showing density of events.

Visual: A line graph representing density of mouse clicks by participants across tasks over time.

Physiological Techniques

Eye Tracking

Eye movement tracking shows where participants focus on a screen, in what order.

Gaze patterns reveal how people move around the screen. **Heatmaps** highlight most looked-at areas (red = most, yellow = fewer, blue = fewest, gray = not viewed).

Technology

- Head-mounted or portable eye-tracking devices.
- Records where on the screen the participant is looking.

Visual: A heatmap overlaid on a webpage showing F-shaped scanning pattern – users prioritize top and left areas, focusing on titles and descriptions.

Visual: Heatmap on a search results page showing red-hot areas around top results.

Benefits

- Indicates which features attracted attention.
- Helps understand scanning behavior.

Other Physiological Measures

- **Heart rate**
- **Perspiration**
- **Skin temperature**
- These measure changes in **arousal** as a result of emotional response.
- **Problem:** The specific emotion evoked cannot be identified.

Review-based and Model-based Evaluation

- **Review-based evaluation:** Results from literature used to support or refute parts of design. Care needed to ensure transferability.
 - **Model-based evaluation:** Cognitive models (e.g., GOMS) used to filter design options and predict user performance. Design rationale can also provide useful evaluation information.
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Key Takeaways from Test Questions

- **Menu-based interfaces** eliminate the need to remember commands.
- **Natural language** relieves the burden of learning syntax (e.g., typing on WhatsApp).
- **Gestural interfaces** face challenges like gesture ambiguity and fatigue.
- **Heuristic evaluation** is based on expert judgement using heuristics.
- Severity in heuristic evaluation depends on frequency, impact, persistence – **not usability** itself.
- **Formative evaluation** is conducted early, during requirements gathering.
- **Colour** can distinguish interactive elements from non-interactive ones.
- **Meaning ambiguity** is not a benefit of multimodal interfaces (it is a challenge).
- Heuristic evaluation checks for violations like insufficient feedback, unattractive design, inconsistency, unpredictability.
- **QWERTY keyboard** layout has the first six keys on top row.
- **Tangible interface** allows interaction with virtual objects via physical objects.
- **Gestural interaction** interprets human gestures via mathematical algorithms.
- Voice assistants (Siri, Alexa) are examples of **natural language** interaction.
- **Summative evaluation** is carried out at the end of the lifecycle.
- **Direct manipulation** attributes include easy learning by novices, fast operation by experts, less anxiety; it does **not** make the interface incomprehensible.
- Adaptive interfaces track user interaction and predict tasks.
- Icons in menus provide **visual cues for faster recognition**.
- **Modality** is the sense by which a message is sent or received.

- Speech interaction suits hands-free, natural language in noisy environments.
- Combining visuals and sound reduces load on visual system and frees screen space.
- Gestural interfaces perform better in noisy environments.
- Multimodal interfaces can combine gesture, auditory, visual, touch, sketching.
- Error prevention includes automatic completion, selection rather than typing, gray out inappropriate actions.
- Speech sound disadvantages: slow, many words, context-dependent – all of the above.
- Inclusive design means creating usable products for everyone, recognizing diversity.
- UI evolution: computer to improve productivity, computer as tools, computer in our everyday lives – **not** to replace humans.
- Tree-structured menu is best for accessing medical records by department.
- Command-line is most suitable for expert users wanting speed.
- Trackball is an **indirect** pointing device.
- Whistling and humming are **non-speech sound**.
- Virtual assistants, chatbots, VUIs all support natural language.
- Multi-item menu examples: editing multiple rows, selecting multiple files, filtering by multiple criteria.
- Adaptable interfaces enhance user satisfaction.
- GUI is based on **direct manipulation**.
- Patchy design from adapting existing designs partially addresses inclusivity.
- Survey mode choice does **not** depend on quality of experts who evaluate the survey.
- Voice input is commonly combined with gestural interfaces.
- Speech input is **not** a visual modality.
- Questionnaires advantage: fixed responses with alternatives, scale responses, full coverage – all of the mentioned.
- Binary menu: confirm file delete (Yes/No).
- Sound recommended because people cannot see outside view, harder to avoid hearing, faster reaction – all of the above.

- Direct manipulation involves interacting with objects on screen directly.
- Multimodal interfaces involve only one modality is **false**.
- Subjective satisfaction is usually measured by interviews and surveys.
- Usability inspection methods: Cognitive Walkthrough and Heuristic Evaluation – all of the above.
- Pull-down menu groups related items.

The notes above cover all major topics from the provided source content, organized logically with definitions, guidelines, examples, and visual descriptions. Each section includes key terms in **bold** and important distinctions for study. The content ends substantively with the summary of evaluation methods and test takeaways.